3

Sensors, Endpoints, and Power Systems

The **Internet of Things** (**IoT**) begins with sources of data or devices that perform an action. These we call **endpoints** or **nodes**, and they are the things associated with the Internet. When one discusses IoT, in general, the actual sources of data are often overlooked. These sources are sensors outputting a stream of time-correlated data that must be transmitted securely, possibly analyzed, and possibly stored. The value of IoT is in the data in aggregate. Therefore, the data a sensor provides is crucial. However, for an architect, it is critical to understand the data as well as how the data is interpreted. Besides understanding what data is collected and how it is acquired, in a massive IoT deployment, it is useful to know what can be sensed, and what the constraints are for various sensors. For example, a system must account for lost devices and erroneous data. An architect must understand the reasons why data may not be reliable from sensors, and how a sensor can fail in the field. Essentially, we are connecting the analog world to the digital. The majority of connected things will be sensors, so it is important to understand their role.

This is what the IoT is in a nutshell. As we learned in *Chapter 1*, *IoT and Edge Computing Definition and Use Cases*, sensors and edge devices comprise most of the growth in IoT, so it is important to understand their relationship in the architecture. This chapter will highlight sensor devices from an electronics and system point of view. It is important to understand the principles of what is being measured, and why.

One should ask, "What type of sensor or edge device should I consider for the problem I'm trying to solve?" An architect should consider aspects of cost, features, size, usable life, and precision when deploying an IoT solution. Additionally, the power and energy of edge devices are rarely addressed in IoT literature but are critical to building reliable and long-lasting technology. The reader should leave this chapter with a high-level grasp of sensor technology and its constraints.

In this chapter, we will cover the following topics:

- Sensing devices from thermocouples to MEMS sensors to vision systems
- Energy generation systems
- Energy storage systems

Sensing devices

We begin by focusing on the sensing or input devices. These come in a variety of forms and complexities, from simple thermocouples to advanced video systems. One reason IoT is a large growth area is the fact that these sensing systems have been reduced in size and cost substantially, with advances in semiconductor fabrication and micromachining.

Thermocouples and temperature sensing

Temperature sensors are the most prevalent form of sensor products. They exist just about everywhere. From smart thermostats to IoT cold storage logistics, from refrigerators to industrial machinery, they are prevalent and most likely the first sensing device you will be exposed to in an IoT solution.

Thermocouples

A **thermocouple** (or **TC**) is a form of temperature-sensing apparatus that doesn't rely on an excitation signal to operate. Therefore, they produce very small signals (often microvolts in amplitude). Two wires of two different materials meet where a temperature measurement is to be sampled. Each metal develops a voltage differential independently of each other. This effect is known as the **Seebeck electromotive effect**, in which the difference between the voltage of the two metals has a nonlinear relationship to the temperature.

The magnitude of the voltage depends on the metal material chosen. It is critical that the ends of the wires are thermally isolated from the system (and the wires need to be at the same controlled temperature). In the following diagram, you will see a thermal block whose temperature is controlled with a sensor. This is usually controlled via a technique called **cold junction compensation**, where the temperature can vary, but is measured accurately by the block sensor.

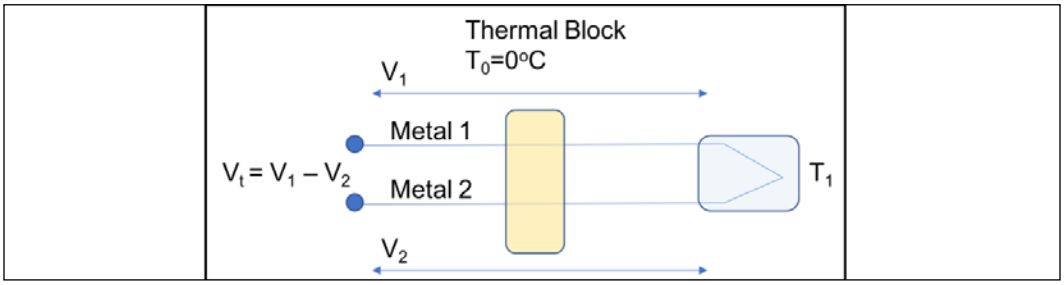


Figure 1: Thermocouple schematic

When sampling the voltage differential, the software will usually have a lookup table to derive the temperature based on the nonlinear relationship of the metals chosen.

Thermocouples should be used for simple measurements. The system can vary in accuracy as well, since subtle impurities can affect the wire composition and cause a mismatch in the lookup tables. Precision-grade thermocouples may be needed but come at a higher cost.

Another effect is aging. Since thermocouples are often used in industrial settings, high-heat environments can degrade the accuracy sensors over time. Therefore, IoT solutions must account for changes over the life of a sensor.

Thermocouples are good for wide temperature ranges, color-coded for different metal combinations, and labeled by type (for example, E, M and PT-PD, to name a few). In general, these sensors are good for long distance measurements with long leads and often used in industrial and high-temperature environments.

Shown in the following figure are various thermocouple metal types and their respective energy linearities across a range of temperatures.

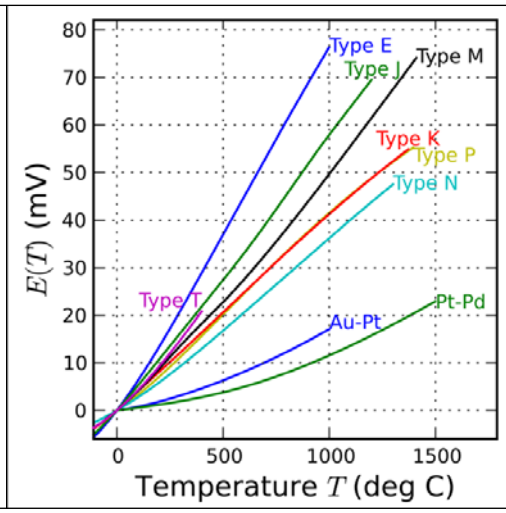


Figure 2: Thermocouple type characterization E(T):T

Resistance temperature detectors

Resistance temperature detectors (**RTDs**) operate within a narrow range of temperatures but have much better accuracy than thermocouples (below 600 degrees Celsius). These are usually constructed with very fine platinum wire tightly wrapped around ceramic or glass. This produces a resistance-to-temperature relationship. Because it is a resistance-based measurement, an excitation current is needed to operate an RTD (1 mA).

The resistance of an RTD follows a predefined slope. RTDs are specified with a base resistance. A 200 PT100 RTD has a slope of 0.00200 Ohms/degrees Celsius from 0 to 100 degrees Celsius. Within that range (0 to 100 degrees Celsius), the slope will be linear. RTDs come in two-, three-, and four-wire packages, with four-wire models used for high-accuracy calibration systems. RTDs are often used with bridge circuits to increase resolution, with software linearizing the results. The following illustration shows the wire-round RTD design and form factor.

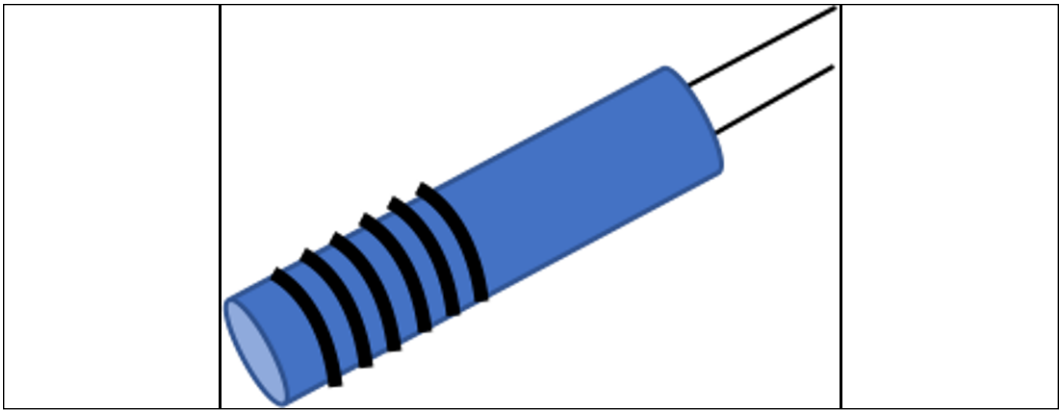


Figure 3: Wire-wound RTD

RTDs are rarely used above 600 degrees Celsius, which limits their application in industry. At high temperatures, the platinum can become contaminated, leading to incorrect results; however, when measuring within their specified range, RTDs are fairly stable and accurate.

Thermistors

The last temperature sensing device is the **thermistor**. These are also resistance-based relationship sensors like RTDs but produce a higher degree of change for a given temperature than an RTD. Essentially, these are resistors that vary based on temperature. They are also used in circuits to alleviate in-rush currents. Whereas an RTD has a linear relationship with a change in temperature, thermistors have a highly nonlinear relationship and are suitable where a high resolution is needed for a narrow temperature range. There are two types of thermistors: NTC, where resistance decreases as temperatures rise; and PTC, where resistance increases with rising temperatures. The main difference from an RTD materially is the use of ceramics or polymers, whereas metals are the basis of RTDs.

Thermistors are found in medical devices, scientific equipment, food handling equipment, incubators, and home appliances such as thermostats.

Temperature sensor summary

In summary, the following table highlights the use cases and benefits of particular temperature sensors:

Category	**Thermocouples**	**RTDs**	**Thermistors**
Temperature range (degrees Celsius)	-180 to 2,320	-200 to 500	-90 to 130
Response time	Fast (microseconds)	Slow (seconds)	Slow (seconds)
Size	Large (~1 mm)	Small (5 mm)	Small (5 mm)
Accuracy	Low	Medium	Very high

Typically, a thermocouple will be used for measuring temperatures in industrial applications and consumer heating, such as a kiln or furnace. These require fast response times from electronics to make real-time adjustments in extreme temperature environments. Thermistors are used for general purpose temperature sensing, such as garden-variety Bluetooth or Zigbee sensors, digital thermometers, electric motors, fire alarms, cold storage, and refrigerators.

Hall effect sensors and current sensors

A **Hall effect sensor** consists of a strip of metal that a current runs across. A stream of charged particles passing through a magnetic field will cause the beam to deflect from a straight line. If a conductor is placed in the magnetic field perpendicular to the electron flow, it will gather charge carriers and produce a voltage differential between the positive side of the metal strip and the negative side. This will produce a voltage differential that can be measured. The differential is called the **Hall voltage**, which is responsible for the phenomenon known as the **Hall effect**. This is illustrated in the following image. If a current is applied to a metal strip (as shown in the diagram) within a magnetic field, electrons will be attracted to one side of the strip, and holes to the other (see the curved line). This will induce an electric field that can be measured. If the field is strong enough, it will cancel out the magnetic force, and electrons will follow the straight line.

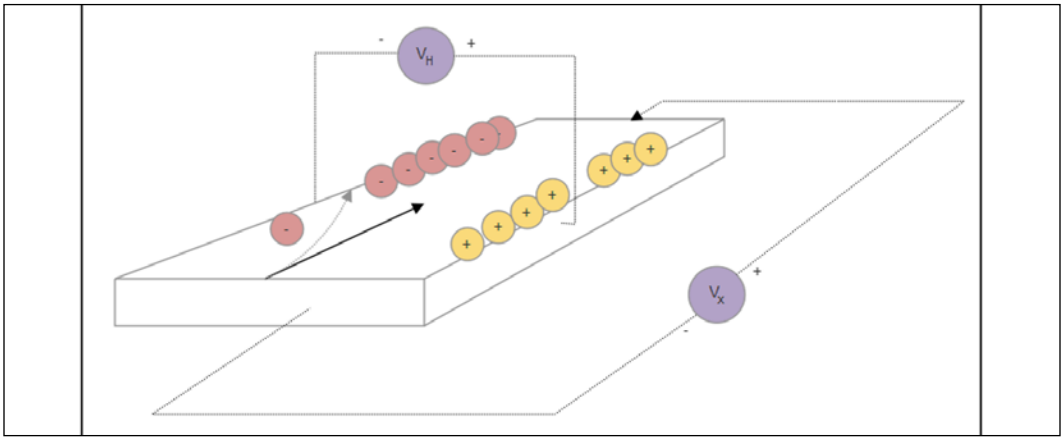


Figure 4: Example of the Hall effect

Current sensors use the Hall effect for measuring the AC and DC currents of a system. There are two forms of current sensors: open loop and closed loop. Closed loops are more expensive than open loop sensors and often used in battery-powered circuits.

Typical uses of Hall sensors include position sensing, magnetometers, highly reliable switches, and water-level detection. They are used in industrial sensors to measure the rotational velocity of different machines and motors. Additionally, these devices can be created very inexpensively and tolerate harsh environmental conditions.

Photoelectric sensors

The detection of light and light intensity is used in many IoT sensor devices, such as security systems, smart switches, and smart street lighting. As the name implies, a photoresistor varies in resistance depending on light intensity, while photodiodes convert light into an electrical current.

Photoresistors are manufactured using a high-resistance semiconductor. The resistance decreases as more light is absorbed. In the dark, a photoresistor can have quite a high resistance (in the megaohm range). Photons absorbed by the semiconductor allow electrons to jump to the conduction band and conduct electricity. Photoresistors are wavelength sensitive, depending on their type and manufacturer; however, photodiodes are true semiconductors with a *p-n* junction. The device responds to light by creating an electron-hole pair.

The holes move toward an anode, the electrons migrate to a cathode, and a current is produced. Traditional solar cells operate in this photovoltaic mode, producing electricity. Alternatively, a reverse bias can be used on the cathode to improve the latency and response time if needed. The following table contrasts photoresistors with photodiodes.

Category	**Photoresistor**	**Photodiode**
Light sensitivity	Low	High
Active/passive (semiconductor)	Passive	Active
Temperature sensitivity	Highly sensitive	Low
Latency to light changes	Long (10 milliseconds on, 1 second off)	Short

PIR sensors

Pyroelectric infrared (**PIR**) sensors contain two slots filled with material that reacts to IR radiation and heat. Typical use cases are security or warm-body movement. In its simplest form, a Fresnel lens sits on top of the PIR sensor, allowing both slots to form a widening arc outward. These two arcs create detection zones. When a warm body enters one of the arcs, or leaves one, it generates a signal that is sampled. The PIR sensors use a crystalline material that generates current when subjected to IR radiation. A **field-effect transistor** (**FET**) detects the change in current and sends the signal to an amplification unit. PIR sensors respond well in the 8 to 14 um range, which is typical of a human body.

The following diagram illustrates two IR regions detecting two zones. While this is fine for some purposes, generally we need to inspect an entire room or area for movement or activity:

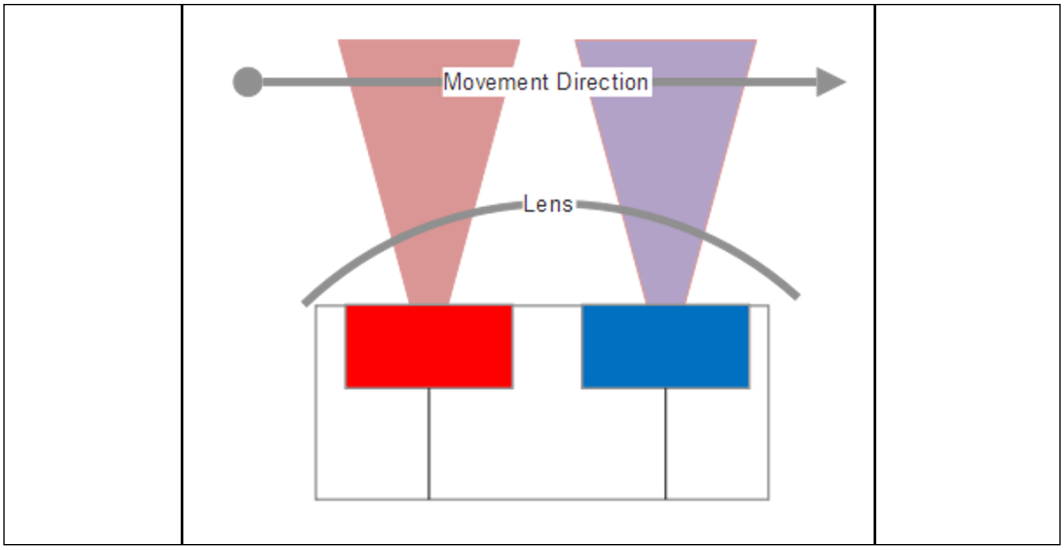


Figure 5: PIR sensor: Two elements responding to a source of IR moving across its field of view

To scan a larger area with a single sensor requires multiple Fresnel lenses that condense light from regions of the room to create distinct regions on the PIR array. This also has the effect of condensing the infrared energy to discrete FET areas. Typically, such devices allow the architect to control the sensitivity (range) as well as the hold time.

The hold time specifies for how long to output a motion event after an object has been detected moving across the path of the PIR. The shorter the hold time, the more events may be output.

The following diagram shows a typical PIR sensor with a Fresnel lens focusing on the substrate by a fixed focal length:

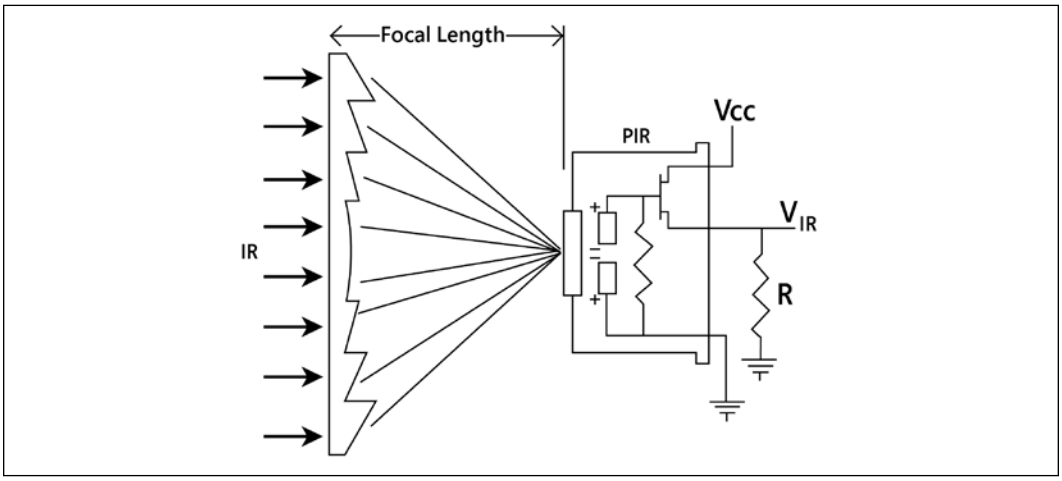


Figure 6: Left: Fresnel lens condensing IR region onto PIR sensor

LiDAR and active sensing systems

This section will cover active sensing systems. We have discussed many passive sensors that simply respond to environmental changes. Active sensing involves broadcasting a signal that is referenced to measure the environment either spatially or temporally. While this area is broad, we will concentrate on LiDAR as a basis for active sensing systems.

LiDAR stands for **Light Detection and Ranging**. This type of sensor measures the distance to a target by measuring a laser pulse reflection on the target. Where a PIR sensor will detect movement within a range, LiDAR is capable of measuring a range. The process was first demonstrated in the 1960s and is now used pervasively in agriculture, automated and self-driving vehicles, robotics, surveillance, and environmental studies. This type of active sensing machine is also capable of analyzing anything that crosses its path. They are used to analyze gases, atmospheres, cloud formations and compositions, particulates, the speed of moving objects, and so on.

LiDAR is an active sensor technology and broadcasts laser energy. As a laser hits an object, some of the energy will be reflected back towards the LiDAR emitter. The lasers used are typically in the 600 to 1000 nm wavelength and relatively inexpensive. Power is constrained for safety reasons to prevent eye damage. Some LiDAR units operate in the 1550 nm range, as that wavelength cannot be focused by the eye, rendering them harmless even at high energy.

LiDAR systems are capable of very long ranging and scanning, even from satellites. The laser will pulse up to 150,000 pulses per second, which can reflect off an object back to a photodiode array. The laser apparatus can also sweep the scene via a rotating mirror to build a comprehensive 3D image of the environment. Each beam that is broadcast represents an angle, **time-of-flight** (**ToF**) measurement, and a GPS location. This allows the beams to form a representative scene.

To calculate the distance to an object, the equation is relatively simple:

$$Distance = \frac{(Speed\ of\ Light \times Time\ of\ Flight)}{2}$$

LiDAR and other active sensors behave in a similar manner. Each has a representative broadcasting signal that returns to a sensor to build an image or indicate that an event has occurred. These sensors are much more complex than simple passive sensors and also take more power, cost, and area.

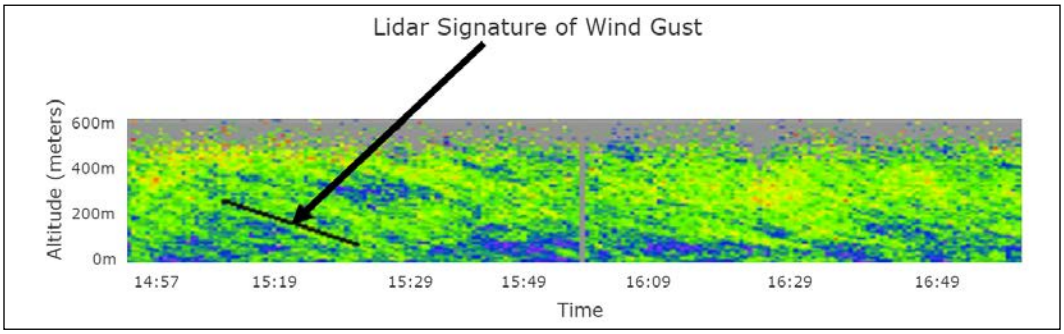


Figure 7: LiDAR: An example of a LiDAR image used to analyze atmospheric wind gusts to protect wind turbines. Blue areas signify wind speeds of 0 to 2m/s. Green areas show wind speeds of 3 to 6m/s. Red line indicates wind gusts of 7+ m/s. Image courtesy of NASA.

MEMS sensors

Microelectromechanical systems (**MEMS**) have been in the industry since they were first produced in the 1980s; however, the origin of the first MEMS pressure sensor dates back to the 1960s at Kulite Semiconductor, which developed a piezoresistive pressure sensor.

Essentially, they incorporate miniaturized mechanical structures that interact with electronic controls. Typically, these sensors are in the 1 to 100 um geometry range. Unlike the other sensors mentioned in this chapter, MEMS mechanical structures can spin, stretch, bend, move, or alter form, which in turn affects an electrical signal. This is the signal captured and measured by one particular sensor.

MEMS devices are manufactured in a typical silicon fabrication process using multiple masks, lithography, deposition, and etching processes. MEMS silicon dies are then packaged with other components such as operational amplifiers, analog to digital converters, and support circuitry. Typically, MEMS devices will be fabricated in the relatively large 1- to 100-micron range, whereas typical silicon structures are fabricated at 28 nm or below. The process involves a thin layer deposition and etching to create the 3D structures for a MEMS device.

Besides sensor systems, MEMS devices can be found in the heads of inkjet printers and modern overhead projectors such as **digital light processor** (**DLP**) projectors. The ability to synthesize MEMS sensing devices into packages as small as a pinhead has and will allow IoT growth into billions of connected things.

MEMS accelerometers and gyroscopes

Accelerometers and gyroscopes are common in many mobile devices today and used in positioning and movement tracking, such as with pedometers and fitness trackers. These devices will use a MEMS piezoelectric to produce a voltage in response to movement.

Gyroscopes detect rotational motion, and accelerometers respond to changes in linear motion. The following diagram illustrates the basic principle of an accelerometer. Typically, a central mass that is fixed to a calibrated location via a spring will respond to changes in acceleration that are measured by varying capacitance in a MEMS circuit. The central mass will appear to be stationary in response to acceleration in one direction.

An accelerometer will be synthesized to respond to multiple dimensions (*X*, *Y*, *Z*) rather than one dimension, as shown in this illustration:

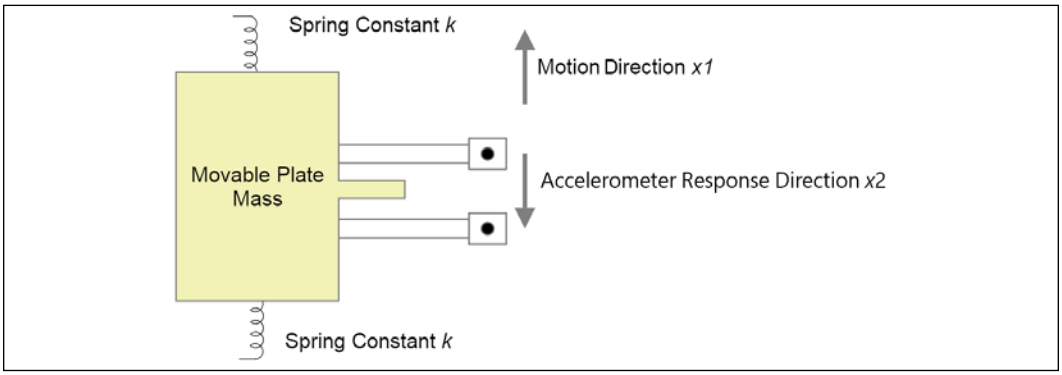


Figure 8: Accelerometer: The principle of acceleration measurement using a central mass suspended by a spring. Typically, these will be used in multiple dimensions.

Gyroscopes operate slightly differently. Rather than relying on the motion response to a central mass, gyros rely on the Coriolis effect of a rotating reference frame. The following figure demonstrates the concept. Without increasing velocity, the object would move in an arc and not reach the northbound target. Moving towards the outer edge of the disk requires additional acceleration to maintain a northbound course. This is the Coriolis acceleration. In a MEMS device, there is no spinning disk; rather, there is a resonant frequency applied to a series of MEMS-fabricated rings on a silicon substrate:

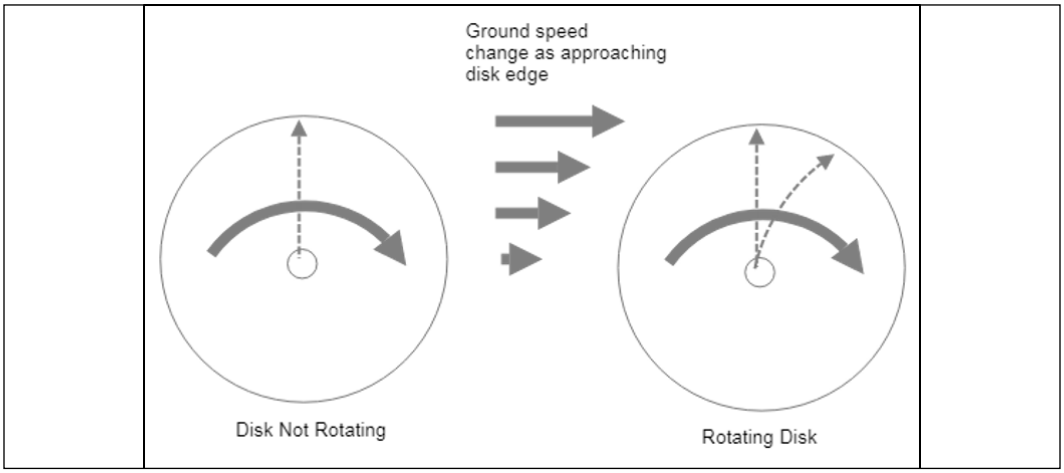


Figure 9: Accelerometer: Effect of a rotating disk on a path moving northward

The rings are concentric and cut into small arcs. Concentric rings allow for more area to gauge the accuracy of rotational movement. Single rings would require rigid support beams and are not as reliable. By bifurcating the rings into arcs, the structure loses rigidity and is more sensitive to rotational forces. The DC source creates an electrostatic force that resonates within the ring, while the electrodes attached to the rings detect the changes in the capacitor. If the resonating rings are perturbed, the Coriolis acceleration is detected. The Coriolis acceleration is defined by the following equation:

$$a = -2\omega \times v$$

This equation states that the acceleration is a product of the rotation of the system and the velocity of the rotating disk, as shown in the preceding diagram, or the resonant frequency of the MEMS device, as shown in the following diagram. Given a DC power source, a force changes the gap size and the overall capacitance of the circuit.

The outer electrodes detect deflection in the ring, while inner electrodes provide capacitance measurements:

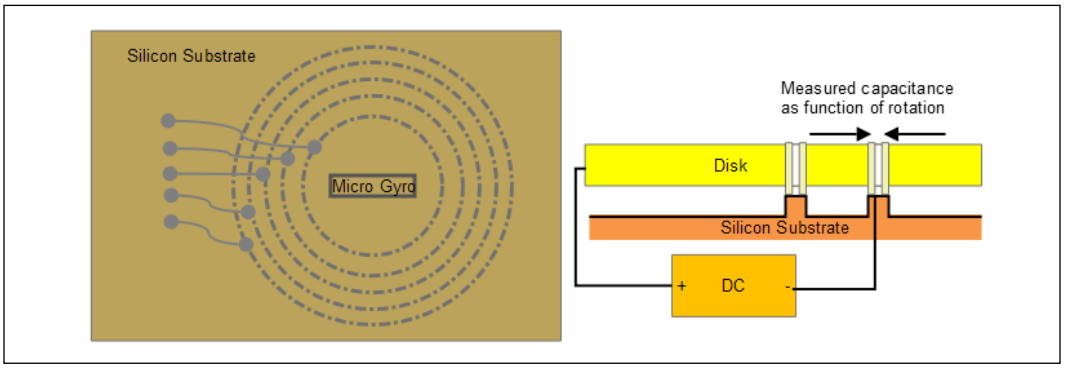


Figure 10: Left: Concentric cut rings representing the gyro sensor sitting upon a silicon substrate. Right: The disk gap connected to a corresponding silicon substrate.

Both gyroscopes and accelerometers will require power supplies and an op-amp for signal conditioning. After conditioning, the output is ready to be sampled by a digital signal processor.

These devices can be synthesized in very small packages, such as the InvenSense MPU-6050, which includes a six-axis gyro and accelerometer in a small 4 mm x 4 mm x 1 mm package.

The device draws on 3.9 mA of current and is good for low power sensing.

A device such as the InvenSense MPU-6050 is typically used in gaming consoles, smart TVs, and smartphones. This allows a user to have an interactive and engaging experience with the appliance. A device like this can accurately track user motions in a three-dimensional space.

MEMS microphones

MEMS devices can also be used for sound and vibration detection. These types of MEMS devices are related to the accelerometers previously covered. For IoT deployments, sound and vibration measurements are common in the industrial IoT and predictive maintenance applications. For example, an industrial machine that rotates or mixes a load of material in chemical manufacturing, or in centrifuges to separate mixtures, needs precise leveling. A MEMS sound or vibration unit will typically be used to monitor the health and safety of such equipment.

This type of sensor will require an analog-to-digital converter of sufficient sampling frequency. Additionally, an amplifier is used to strengthen the signal.

The impedance of a MEMS microphone is in the order of several hundred ohms (which requires careful attention to the amplifier used). A MEMS microphone can be analog or digital. An analog variety will be biased to some DC voltage and attached to a codec for analog-to-digital conversion. A digital microphone has the ADC close to the microphone source. This is useful when there is signal interference from cellular or Wi-Fi signals near the codec.

The output of a digital MEMS microphone can either be **pulse density modulated** (**PDM**) or sent over I^2S format. PDM is a high sampling rate protocol that has the ability to sample from two microphone channels. It does this by sharing a clock and data line, and sampling from one of the two microphones on different clock periods. I^2S doesn't have a high sampling rate, and the decimation at audio rates (Hz to kHz range) results in fair quality. This still allows multiple microphones to be used in sampling, but may not require an ADC at all, since the decimation is happening in the microphone. A PDM, with its high sampling rate, will need to be decimated by a **digital signal processor**, or **DSP**.

MEMS pressure sensors

Pressure and strain gauges are used in a variety of IoT deployments, from smart cities monitoring infrastructure to industrial manufacturing. These are typically used to measure fluid and gas pressures. The heart of the sensor is a piezoelectric circuit. A diaphragm will be placed above or below a cavity on the piezoelectric substrate. The substrate is flexible and allows the piezo crystals to change shape. This change in shape results in a directly correlated resistance change in the material.

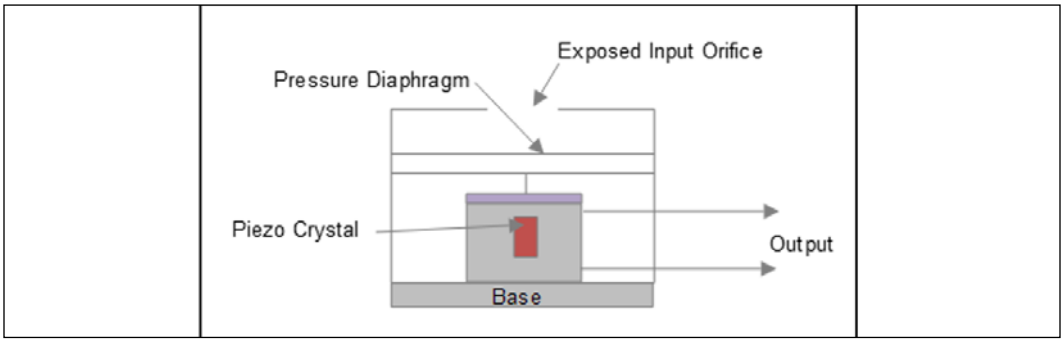


Figure 11: Pressure sensor anatomy

This type of sensor, as well as others listed in this chapter based on an excitation current, relies on a **Wheatstone bridge** to measure changes. Wheatstone bridges may come in two-, four-, or six-wire combinations.

The change in voltage is measured across the bridge as the piezoelectric substrate flexes and changes resistance:

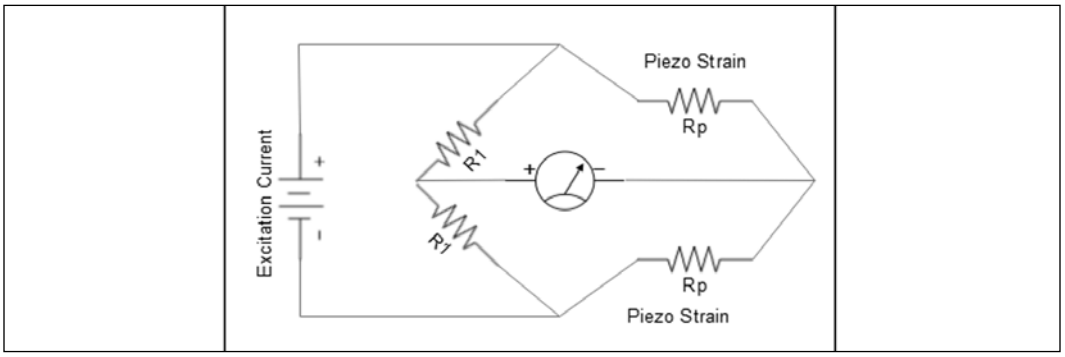


Figure 12: Wheatstone bridge used for amplification of MEMS pressure sensor

High performance IoT endpoints

So far, we have examined simple sensors that simply return information in a binary or analog form that must be sampled. There are, however, IoT devices and sensors that have substantial processing power and performance for the tasks they undertake. Smart sensors include devices such as video cameras and vision systems. Smart sensors can include substantial amounts of processing capacity in the form of high-end processors, digital signal processors, FPGAs, and custom ASICs. In this section, we will explore the details of one form of smart sensor: a **vision system**.

Vision systems

In contrast with the simple sensors explored earlier, vision systems are much more complex, which results in substantial hardware, optics, and imaging silicon. Vision systems start with a lens that observes a scene. A lens provides focus, but also provides more light saturation to the sensing element. In modern vision systems, one of two types of sensing elements is used: **charge-coupled devices** (**CCDs**), or **complementary metal-oxide semiconductor** (**CMOS**) devices. The difference between CMOSes and CCDs can be generalized as follows:

- **CCD**: Charge is transported from the sensor to the edge of the chip to be sampled sequentially via an analog-to-digital converter. CCDs create high-resolution and low-noise images. They consume considerable power (100x that of CMOS). They also require a unique manufacturing process.

- **CMOS**: Individual pixels contain transistors to sample the charge and allow each pixel to be read individually. CMOS is more susceptible to noise but uses little power.

Most sensors are built using CMOS in today's market. A CMOS sensor is integrated into a silicon die that appears as a two-dimensional array of transistors arranged in rows and columns over a silicon substrate. A series of microlenses will sit upon each red, green, or blue sensor focusing incidental rays onto transistor elements. Each of these microlenses attenuates a specific color to a specific set of photodiodes (*R*, *G*, or *B*) that respond to the level of light; however, lenses are not perfect. They can add chromatic aberrations where different wavelengths refract at different rates, which leads to different focal lengths and blur. A lens can also distort an image, causing pincushioning effects.

Next will come a series of steps to filter, normalize, and convert the image several times into a usable digital image. This is the heart of the **image signal processor** (**ISP**), and the steps may be followed in this order:

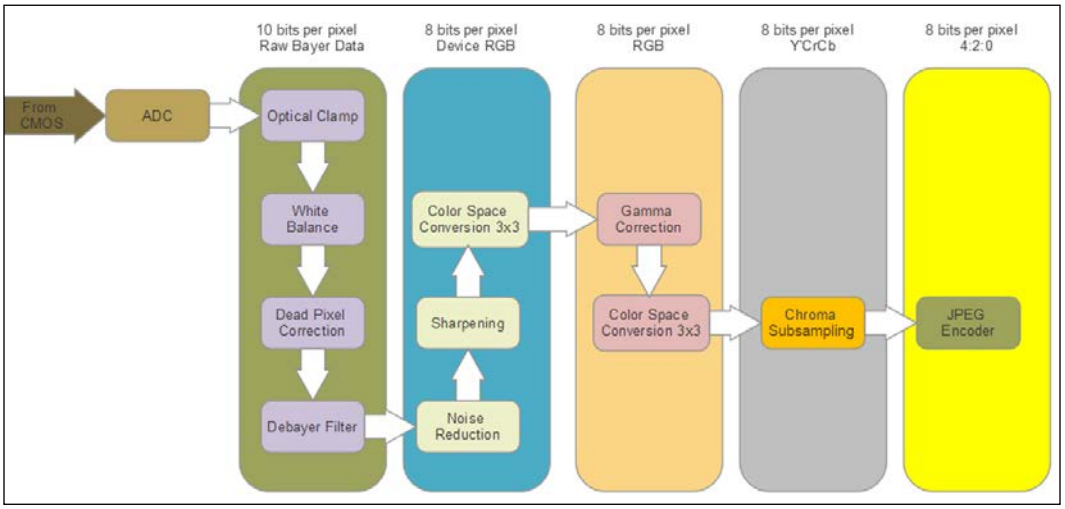


Figure 13: Image sensor: Typical image signal processor pipeline for color video

Note the numerous conversions and processing of each stage of the pipeline for each pixel in the image. The amount of data and processing requires substantial custom silicon or digital signal processors. The following lists the functional block responsibilities in the pipeline:

- **Analog-to-digital conversion**: Amplification of sensor signal then converted to digital form (10-bit). Data is read in from the photodiode sensor array as a flattened series of rows/columns representing the image just captured.

- **Optical clamp**: Removes sensor biasing effects due to sensor black level.
- **White balance**: Mimics chromatic display in the eye for different color temperatures. Neutral tones appear neutral. This is performed using matrix conversions.
- **Dead pixel correction**: Identifies dead pixels and compensates for their loss using interpolation. Dead pixels are replaced with the average of neighboring pixels.
- **Debayer filtering and demosaicing**: Debayer filtering separates RGB data to saturate green over red and blue content for luminance sensitivity adjustment. It also creates a planar format of images from sensor-interlaced content. More advanced algorithms preserve edges in images.
- **Noise reduction**: Sensors introduce noise due to aliasing effects, analog-to-digital conversions, and so on. Noise can be associated with the non-uniformity of pixel sensitivity at a transistor level, or leakage of the photodiode, revealing dark regions. Other forms of noise exist as well. This phase removes white and coherent noise introduced in the image capture through a median filter (3 x 3 array) across all pixels. Alternatively, a despeckle filter can be used, requiring pixels to be sorted. Other methods exist as well. However, they all walk the pixel matrix.
- **Sharpening**: Applies a blur to an image using a matrix multiplication, and then combines the blur with detail in content regions to create a sharpening effect.
- **Color space conversion 3 x 3**: Color space conversion to RGB data for RGB-particular treatments.
- **Gamma correction**: Corrects for CMOS image sensor nonlinear response on RGB data to different irradiance. Gamma correction uses a **lookup table** (**LUT**) to interpolate and correct an image.
- **Color space conversion 3 x 3**: Additional color space conversion from RGB to Y'CbCr format. YCC was chosen since *Y* can be stored at a higher resolution than *CbCr* without the loss of visual quality. This also uses 4:2:2-bit representation (4 bits *Y*, 2 bits of *Cb*, and 2 bits *Cr*).
- **Chroma subsampling**: Due to nonlinearities in RGB tones, corrects images to mimic other media such as film for tonal matching and quality.
- **JPEG encoder**: Standard JPEG compression algorithm.

It should be emphasized here that this is a good example of how complex a sensor can be, and how much data, hardware, and complexity can be attributed to a simple vision system. The amount of data passing through a vision system or camera at a conservative 60 frames per second at 1080p resolution is massive.

All the phases (except JPEG compression) move through an ISP in fixed-function silicon (as in an ASIC) one cycle at a time. The total amount of data processed is 1.368 GB per second. Accounting for JPEG compression as the last step brings the amount of data to well over 2 GB/sec in processing through custom silicon and CPU/DSP cores. One would never stream raw Bayer image video to the cloud for processing—this work must be performed as close to the video sensor as possible.

Sensor fusion

One aspect that needs consideration with all the sensor devices described in this chapter is the concept of **sensor fusion**. Sensor fusion is the process of combining several different kinds of sensor data to reveal more about context than a single sensor can provide. This is important in the IoT space, since a single thermal sensor has no notion of what causes a rapid temperature change. However, when combined with data from other sensors nearby that look at PIR motion detection and light intensity, an IoT system could discern that a large number of people are congregating in a certain area while the sun is shining and could then make the decision to increase air circulation in a smart building. A simple thermal sensor only records the current temperature value and has no contextual awareness that the heat is rising due to people congregating and sunlight shining.

With time-correlated data from multiple sensors (edge and cloud), processing can make better decisions based on more data. This is one of the reasons that there will be a large influx of data from sensors to the cloud, and this is causing big data growth. As sensors become cheaper and easier to integrate, as with the TI SensorTag, we will see more combined sensing to provide contextual awareness.

There are two modes of sensor fusion:

- **Centralized**: Where raw data is streamed and aggregated to a central service and fusion occurs there (cloud-based, for example)
- **Decentralized**: Where data is correlated at the sensor

The basis of correlating sensor data is usually expressed through the **central limit theorem**, where two sensor measurements, x_1 and x_2, are combined to reveal a correlated measurement, x_3, based on the combined variances. This is simply adding two measures and weighting the sum by the variances:

$$x_3 = (\sigma_1^{-2} + \sigma_2^{-2})^{-1}(\sigma_1^{-2}x_1 + \sigma_2^{-2}x_2)$$

Other sensor fusion methods used are Kalman filters and Bayesian networks.

Chapter 8, *Edge Computing*, will discuss edge computing in depth and illustrate an example of "sensor fusion," which will reinforce that technologies like this have significant resource and processing requirements that extend beyond simple sensor electronics.

Output devices

Output devices in the IoT ecosphere can be just about anything, from a simple LED to a full video system. Other types of output include actuators, stepper motors, speakers and audio systems, industrial valves, and so on. It stands to reason that these devices need various control systems of differing complexity. Depending on the type of output and the use case they serve, it also should be expected that much of the control and processing needs to be located at the edge, or close to the device (versus complete control in the cloud). A video system, for example, can stream data from cloud providers, but needs output hardware and buffering capabilities at the edge.

In general, output systems can require substantial energy to convert to mechanical movement, thermal energy, or even light. A small hobbyist solenoid to control fluid or gas flow can require 9 to 24 VDC and draw 100 mA to operate reliably and produce five Newtons of force. Industrial solenoids operate in hundreds of volts.

Functional examples (putting it all together)

A collection of sensors alone doesn't have much value until the data they collect is transmitted, processed, or acted upon. That processing can be performed by an embedded controller or sent upstream to a cloud. More hardware is needed to build the system. Typically, the sensors will use established IO interface and communication systems, such as I^2C, SPI, UART, SPI, or other low-speed IOs (covered in *Chapter 11*, *Cloud and Fog Topologies*). Other devices, such as video systems, will need much faster IOs to maintain high resolution and fast video frame rates. Such IOs would include MIPI, USB, or even PCI-Express. To communicate wirelessly, the sensors will need to be used with wireless transport hardware like Bluetooth, Zigbee, or 802.11. All this requires extra componentry, which we will cover in this section.

Functional example – TI SensorTag CC2650

The Texas Instruments CC2650 SensorTag is a good example of an IoT sensor module for development, prototyping, and design. SensorTag has the following features and sensors in the package:

- Sensor input
    - Ambient light sensor (TI Light Sensor OPT3001)
    - Infrared temperature sensor (TI Thermopile infrared TMP007)
    - Ambient temperature sensor (TI light sensor OPT3001)
    - Accelerometer (InvenSense MPU-9250)
    - Gyroscope (InvenSense MPU-9250)
    - Magnetometer (Bosch SensorTec BMP280)
    - Altimeter/pressure sensor (Bosch SensorTec BMP280)
    - Humidity sensor (TI HDC1000)
    - MEMS microphone (Knowles SPH0641LU4H)
    - Magnetic sensor (Bosch SensorTec BMP280)
    - Two push-button GPIOs
    - Reed relay (Meder MK24)
- Output devices
    - Buzzer/speaker
    - 2 LEDs
- Communications
    - Bluetooth Low Energy
    - Zigbee
    - 6LoWPAN

This package is powered by a single CR2032 coin cell battery. Finally, the device can be placed into beacon mode (iBeacon) and used as a message broadcaster.

The following is a block diagram of the CC2650 SensorTag module:

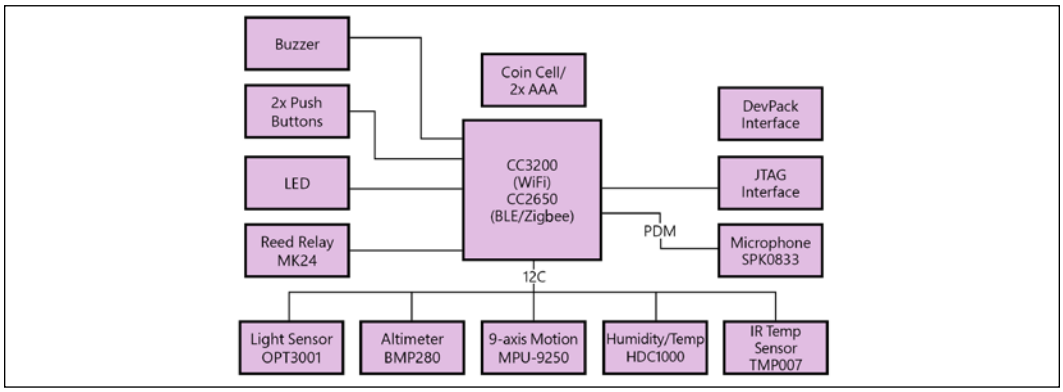


Figure 14: TI CC2650 SensorTag

The next image is of the block diagram of the MCU. The MCU provides the IO and processing ability using an ARM Cortex M3 and ARM Cortex M0, which connect over various bus interfaces to sensor components on the module:

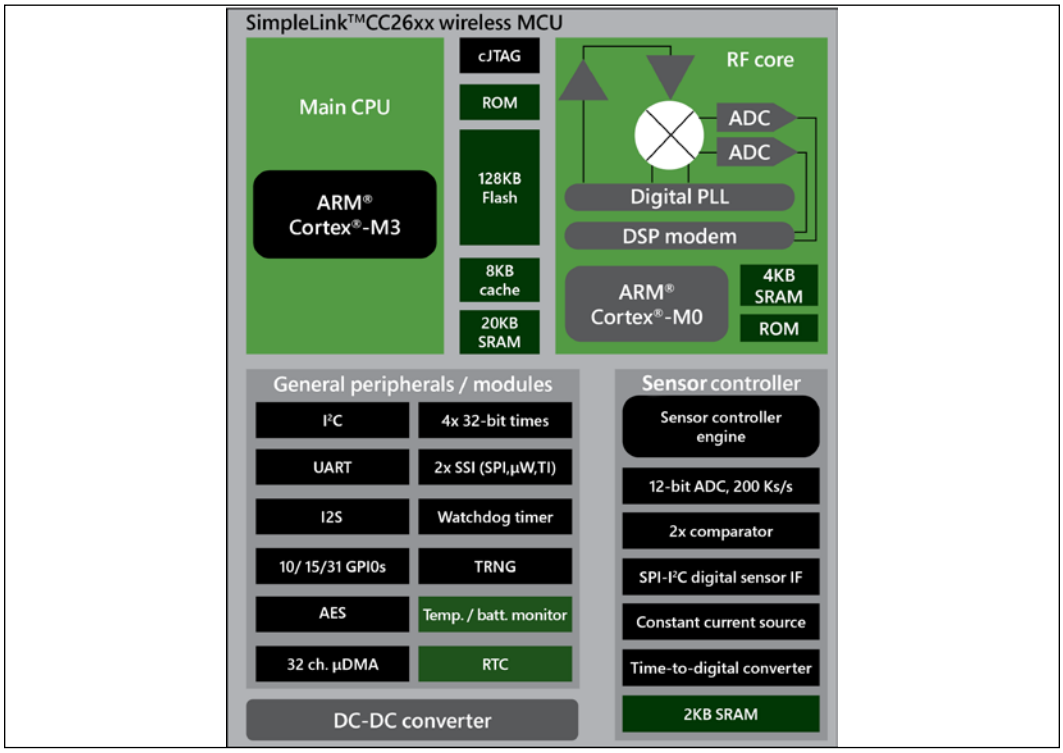


Figure 15: TI CC2650 MCU block diagram

This device is packed with a number of sensors, communication systems, and interfaces, but the amount of processing power is limited. The device uses a processing module from TI (MCU CC265), which includes a small ARM Cortex M3 CPU with only 128 KB of flash memory and 20 KB of SRAM. This was chosen for its extremely low power consumption. The M0 core (notably lower in performance than the M3) manages the radio transmissions.

While power-efficient, this limits the amount of processing and resources on this system. Typically, components like these will need to be accompanied by a gateway, router, cell phone, or some other smart device. Sensor devices such as these that are built for low power and low cost would not have the resources for more demanding applications such as MQTT protocol stacks, data aggregation, cellular communication, or analytics. For that matter, most endpoint sensing devices one will see in the field are simpler than this component to further reduce cost and power.

Sensor to controller

In many of the preceding examples for sensing components, the signal will require amplification, filtering, and calibration before going anywhere. Typically, the hardware will need an analog-to-digital converter of some resolution. The following is a simple 24-bit ADC that outputs a 5V signal:

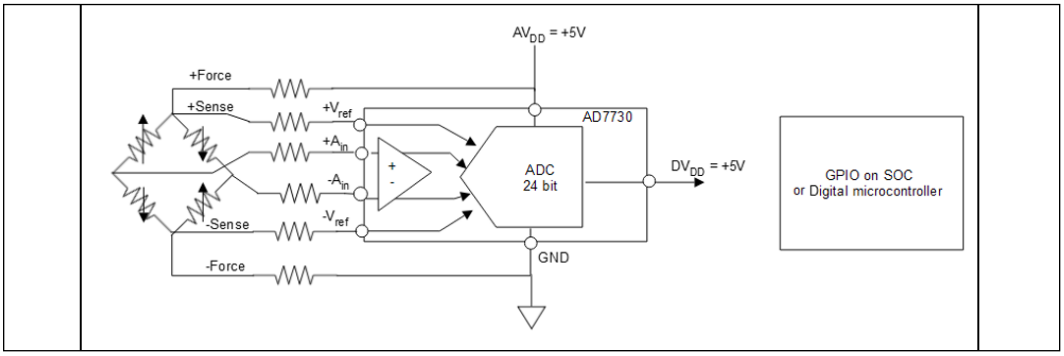


Figure 16: Wheatstone bridge: Connected to an AD7730 analog-to-digital converter as input to a microcontroller or system on a chip

The output can then be raw pulse-modulated data, or a serial interface such as I^2C, SPI, or UART to a microcontroller or digital signal processor. A good example of this in a real system is the Texas Instruments infrared thermopile sensor (TMP007). This is a contactless MEMS temperature sensor that absorbs infrared wavelengths and converts them to a reference voltage while using a cold junction reference temperature.

It is rated for accurately detecting temperatures in environments between -40 degrees Celsius to +125 degrees Celsius. We can see the components covered in this chapter for this part in the following figure:

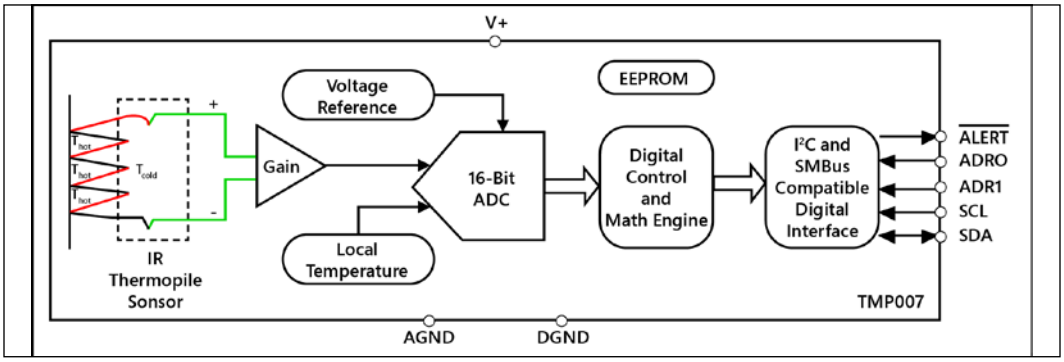


Figure 17: Texas Instruments TMP007 infrared thermopile sensor system

Energy sources and power management

Because sensors and edge devices number in the billions and will be used in very remote areas, delivering reliable power becomes a challenge. Furthermore, there will be IoT deployments where sensors will be buried undersea, or embedded into concrete infrastructures, further complicating power sources. In this section, we will explore the concepts of energy harvesting and power management. Both are very important concepts in the overall IoT story.

Power management

Managing power is a very broad topic and spans software and hardware. It is important to understand the role of power management in a successful IoT deployment, and how to manage power efficiently for remote devices and long-lived devices. The architect must build a power budget for the edge device, which includes:

- Active sensor power
- Frequency of data collection
- Wireless radio communication strength and power

- Frequency of communication
- Microprocessor or microcontroller power as a function of core frequency
- Passive component power
- Energy loss from leakage or power supply inefficiency
- Power reserve for actuators and motors

The budget simply reflects the sum of these power contributors subtracted from the source of power (battery). Batteries also do not have a linear power behavior over time. As the battery loses energy capacity while discharging, the amount of voltage will drop curvilinearly. This poses problems for wireless communication systems. If the battery drops below a minimum voltage, a radio or microprocessor will not reach the threshold voltage and brown out.

For example, the TI SensorTag C2650 has the following power characteristics:

- Standby mode: 0.24 mA
- Running with all sensors disabled (only powering LEDs): 0.33 mA
- Running with all sensors on at 100 ms/sample data rate and broadcasting: 12.08 mA
    - BLE: 5.5 mA
    - Temperature sensor: 0.84 mA
    - Light sensor: 0.56 mA
    - Accelerometer and gyros: 4.68 mA
    - Barometric sensor: 0.5 mA

The TI SensorTag uses a standard CR2032 coin cell battery rated at 240 mAh. Therefore, the maximum life is expected to be about 44 hours. However, the rate of decline changes and is not linear for battery-based devices, as we will see when we cover Peukert's capacity.

Many power management practices are employed, such as clock gating components not being used in silicon, reducing the clock rates of processors or microcontrollers, adjusting the sensing frequency and broadcast frequency, back-off strategies to reduce communication strength, and various levels of sleep modes. These techniques are widely used in the computing business as a general practice.



The techniques described here reflect reactionary power management techniques. They try to minimize energy usage based on dynamic voltage, frequency scaling, and other schemes. New techniques to consider on the horizon include approximate computing and probabilistic design. Both of these schemes rely on the fact that absolute precision is not necessary at all times in a sensor environment running at the edge, especially in use cases involving signal processing and wireless communication. Approximate computing can be done in hardware or software and simply reduces the level of precision in integers when used with functional units such as addresses and multipliers (for example, the value 17,962 is fairly close to 17,970). Probabilistic design realizes many IoT deployments can tolerate certain degrees of faultiness to relax design constraints. Both techniques can reduce the number of gates and power to an almost exponential drop over regular hardware designs.

Energy harvesting

Energy harvesting is not a new concept but is an important concept for IoT. Essentially, any system that represents a change in state (for example, hot to cold, radio signals, light) can convert its form of energy into electrical energy. Some devices use this as their sole form of energy, while others are hybrid systems that use harvesting to augment or extend the life of a battery. In turn, energy harvested can be stored and used (sparingly) to power low-energy devices such as sensors in IoT. Systems must be efficient in capturing energy and storing power. Therefore, advanced power management is needed. For example, if an energy harvesting system uses a piezoelectric mechanical harvesting technique embedded in a sidewalk, it will need to compensate when there is not enough foot traffic to keep the unit charged. Constant communication with energy harvesting systems can further drain the power. Typically, these IoT deployments will use advanced power management techniques to prevent complete loss of functionality.

Techniques such as low standby currents, low-leakage circuits, and clock throttling are frequently used. The following figure illustrates the area where energy harvesting is ideal and the technology that it can power. Care must be taken by the architect to ensure the system is neither underpowered nor overpowered. Harvesting systems, in general, have a low energy potential and low conversion efficiency. An architect should consider energy harvesting in situations where there is a large supply of untapped waste energy, such as in industrial settings.

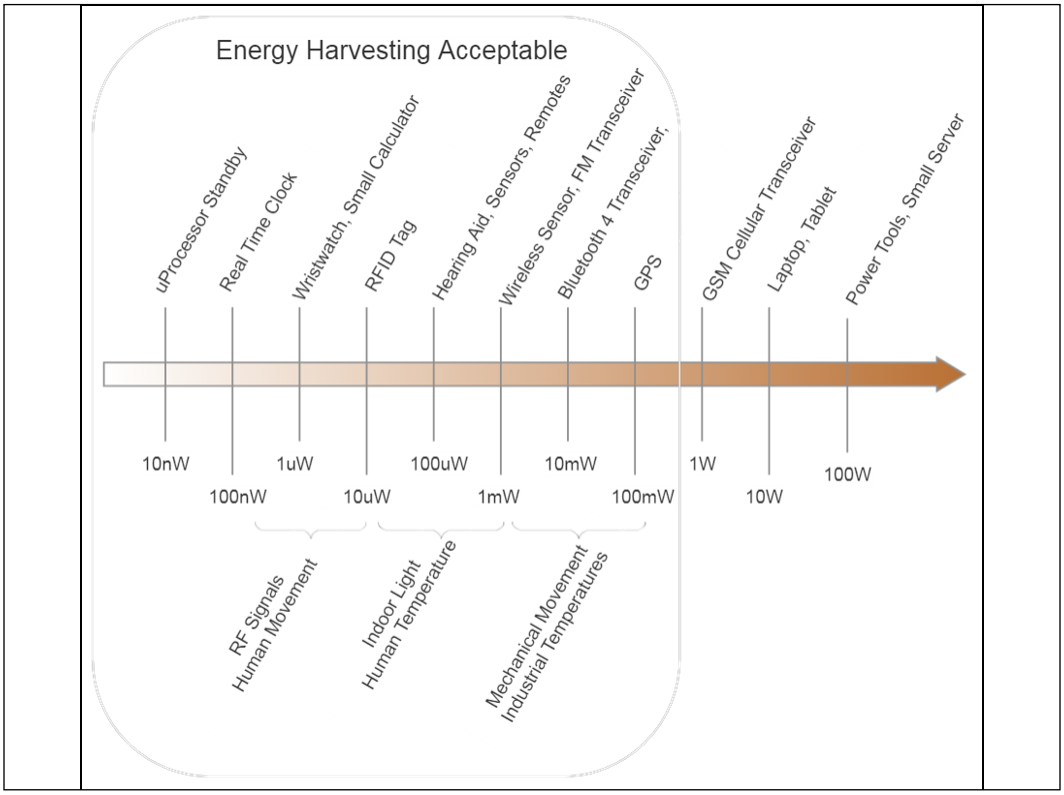


Figure 18: Energy harvesting sweet spots. The figure illustrates typical energy consumption for various devices.

Energy harvesting techniques are common in IoT products for smart cities and remote communication. For example, many cities routinely use traffic count and safety monitors utilizing solar arrays as the main power source. Another example is mail and package drop boxes that use momentary mechanical energy that is captured when the box is opened to harvest the energy to later power electronics monitoring how full the drop box is.

Solar harvesting

Energy from light, whether it is natural or artificial, can be captured and used as an energy source. Earlier in this chapter, we discussed photodiodes and their relationship to sensing light. The same diode can be used in greater quantities to build a traditional solar array. The capacity of energy generation is a function of the area of the solar array. In practice, indoor solar generation is not as efficient as direct sunlight. Panels are rated by their maximum power output in the form of watts.

Solar harvesting is only as effective as how much the sun shines, which varies seasonally and geographically. A region such as the Southwestern US can reclaim considerable energy from direct photovoltaic sources. The Photovoltaic Solar Resource of the United States map was created by the National Renewable Energy Laboratory for the U.S. Department of Energy, `www.nrel.gov` :

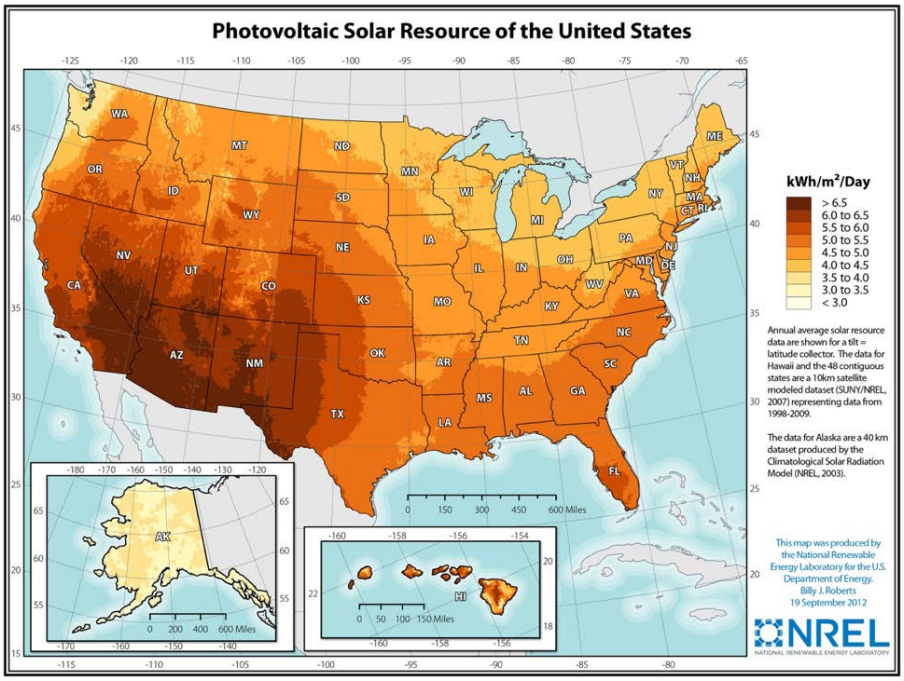


Figure 19: Solar energy map of United States in energy density of kWh/m2 1998-2009

In the United States, Southwest regions fare particularly well with sun intensity, generally lack cloud light barriers, and have good atmospheric conditions, whereas Alaska has the weakest energy density. Solar photovoltaics are not typically efficient. One can expect an 8% to 20% efficiency, with 12% being typical. Regardless, a 25 cm^2 solar array could produce 300 mW at peak power. Another factor is the incidence of light. For a solar collector to achieve such efficiency, the light source must be perpendicular to the array. If the angle of incidence changes as the sun moves, the efficiency drops further. A 12% efficiency collector when the sun is perpendicular will be roughly 9.6% efficient when the sun is 30 degrees from being perpendicular.

The most basic solar collector is the solar cell that is a simple *p-n* semiconductor and similar to the photoelectric sensors discussed earlier. As explained earlier, an electric potential is generated between the *p* and *n* material when a photon is captured.

Piezo-mechanical harvesting

As mentioned earlier in this chapter, piezoelectric effects can be used as sensors, but they can also be used to generate power. Mechanical strains can be converted to energy through motion, vibration, and even sound. These harvesters could be used in smart roadways and infrastructures to harvest and change systems based on traffic movement, even when embedded in concrete. These devices produce currents on the order of milliwatts and thus are suitable for very small systems with some form of energy collection and storage. This process can be performed using MEMS piezo-mechanical devices, electrostatic, and electromagnetic systems.

Electrostatic harvesting incorporates Faraday's law, which basically states that one can induce an electric current by changing the magnetic flux across a coil of wire. Here, the vibration is coupled either to the coil or a magnet. Unfortunately, this scheme in the IoT sensor area provides too little voltage for rectification.

Electrostatic systems use the change in distance between two capacitive plates held at a constant voltage or charge. As the vibration causes the distance to change between the plates, energy (*E*) can be harvested based on the following model:

$$E = \frac{1}{2}QV^2 = \frac{Q^2}{2C}$$

Here, *Q* is the constant charge on the plates, *V* is the constant voltage, and *C* represents capacitance in the preceding equation. Capacitance can also be represented by the length of the plate L_w, the relative static permittivity as ε_0, and the distance between plates *d* as shown:

$$C = \varepsilon_0 L_w / d$$

An electrostatic conversion has the advantage of being scalable and cost-efficient to produce through micromachining and semiconductor fabrication.

The last method for mechanical-to-electrical conversion is **piezo-mechanical**. Piezoelectric devices were talked about earlier in this chapter when discussing sensor input. The same basic concept applies to energy generation. As the piezo-mechanical MEMS device attempts to dampen the mass attached to it, the oscillations will be converted into an electrical current.

Another consideration for the capture and conversion of vibrational or mechanical energy is the need for conditioning before the energy is used or stored. Normally, a passive rectifier is used for conditioning by incorporating a large filtering capacitor. Other forms of energy harvesting do not need such conditioning.

RF energy harvesting

Radiofrequency (**RF**) energy harvesting has been in production for years in the form of RFID tags. RFID enjoys the benefit of being a near-field communication that uses a transceiver that essentially powers the RFID tag due to its close proximity.

For far-field applications, we need to harvest energy from broadcast transmissions. Broadcast transmissions are nearly everywhere, with service from televisions, cell signals, and radio. Capturing energy from radio frequencies versus other forms is particularly difficult, as RF signals have the smallest energy density of all harvesting techniques. The capturing of RF signals is based on having the appropriate antenna to capture a frequency band. Typical frequency bands used are in the 531 to 1611 kHz range (all in the AM radio range).

Thermal harvesting

Thermal energy can be converted into an electric current for any device exhibiting heat flow. Thermal energy can be converted into electrical energy by two basic processes:

- **Thermoelectric**: Direct conversion of thermal energy into electrical energy through the **Seebeck effect**.
- **Thermionic**: Also known as *thermotunneling*. Electrons are ejected from an electrode that is heated and inserted into an electrode that is cool.

The thermoelectric effect (Seebeck effect) is produced when a gradient of temperature exists in conducting materials. The flow of carriers from a hot to a cold region between two dissimilar electrical conductors creates a voltage differential. A thermocouple, or **thermoelectric generator** (**TEG**), could effectively produce voltage simply based on the temperature difference of a human, based on their core body temperature and outside temperature. A temperature difference of 5°C could generate 40 uW at 3V. As heat flows through the conduction material, a hot-side electrode induces electron flow to a cold-side electrode-producing current. Modern thermoelectric devices use *n* or *p*-type bismuth telluride in series. One side is exposed to the source of heat (thermocouple), and the other is isolated.

The energy harvested by the thermopile is proportional to the square of the voltage and equivalent to the temperature difference between the electrodes. One can model the energy harvested by a thermocouple by the following equation:

$$V = \int_{T_L}^{T_H} S_1(T) - S_2(T) dT$$

Here S_1 and S_2 represent the different Seebeck coefficients for each of the two materials (*n* and *p*-type) in the thermopile when there is a temperature differential, T_H -T_L. Since the Seebeck coefficients are functions of temperature and there exists a temperature difference, the result is a voltage difference. This voltage is generally very small, so many thermocouples are used in series to form a thermopile.



The main differences between a thermopile and a thermocouple are the following:

- *Thermocouples* produce a temperature dependent voltage output based on the two conductors forming an electrical connection and measuring the difference of two different temperatures. They are not as accurate for highly precise temperature measurement.
- *Thermopiles* are also electronic devices using many thermocouples, typically in a series circuit. They measure the difference in temperature through the thermal resistance layers. They output voltage proportional to the local temperature difference.

One substantial problem with current thermocouples is the poor efficiency of the energy conversion (less than 10%); however, their advantages are notable, including their small size and ease of manufacturing, resulting in fairly low costs. They also have a very long lifetime of over 100,000 hours. The main problem, of course, is finding a relatively constant source of thermal variance. Using such a device in an environment throughout multiple seasons and temperatures is challenging. For IoT devices, thermoelectric generation typically resides in the 50 mW range.

Thermionic generation is based on the ejection of electrons from a hot electrode to a cold electrode over a potential barrier. The barrier is the work function of the material and best used when there is a significant source of heat energy. While its efficiencies are better than thermoelectric systems, the energy required to jump the potential barrier makes it generally unsuitable for IoT sensor devices. Alternative schemes such as *quantum tunneling* could be considered, but it currently remains a research activity.

Energy storage

Typical storage for an IoT sensor will be a battery or supercapacitor. When considering the architecture for sensor power, one must consider several aspects:

- Volume allowance for a power subsystem. Will a battery even fit?
- The battery energy capacity.
- Accessibility. If the unit is embedded in concrete, limited forms of energy regeneration can be used, and the difficulty of replacing batteries can be costly.
- Weight. Is the unit intended to fly as a drone, or float on water?
- How often will the battery be recharged?
- Is the renewable form of energy constantly available, or intermittent, as in solar?
- The battery power characteristics. How the battery's energy will vary over time as it's discharged.
- Is the sensor in a thermally constrained environment that can affect battery life and reliability?
- Does the battery have a profile that guarantees minimum current availability?

Energy and power models

Battery capacity is measured in amp-hours. A simplified equation to estimate the life of a battery power source is given as:

$$t = \frac{C_p}{I^n}$$

In the equation, C_p is Peukert's capacity, I represents the discharge current, and n is Peukert's exponent. The Peukert effect, as it's known, helps predict the life of a battery where the capacity of a battery decreases at a different rate as discharge increases. The equation shows how discharging at higher rates removes more power from the battery. Alternatively, discharging at lower rates will increase the effective runtime of the battery. One way of thinking of this phenomenon is a battery rated at 100 Ah, and discharging it fully in 20 hours (5A, for this example). If one were to discharge it faster (say, in 10 hours), the capacity would be lower. If one were to discharge it more slowly (say, over 40 hours), it would be greater. However, when represented on a graph, the relationship is nonlinear. Peukert's exponent usually lies between 1.1 and 1.3. As n increases, we move further from a perfect battery to one that discharges faster as the current increases. Peukert's curve applies to lead-acid battery performance, and an example is shown in this next graph:

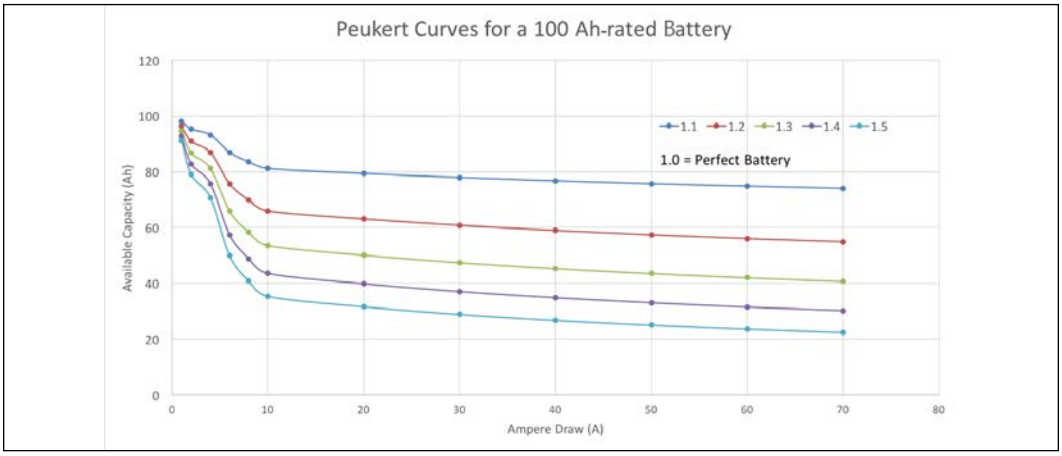


Figure 20: Peukert curves from 1.1 to 1.5 for a battery rated at 100 Ah over 20 hours. The curves show the declining capacity as the Peukert coefficient grows.

One can see the difference in discharge rates for various types of batteries. One advantage of alkaline batteries is that the discharge rate is almost linear in a large portion of the graph. Lithium-ion has a stair-step function in performance, thus making battery charge predictions more difficult. That said, Li-ion provides a near steady and continuous voltage level over the life of the charge, and consistently powers electronics over the span of its charge:

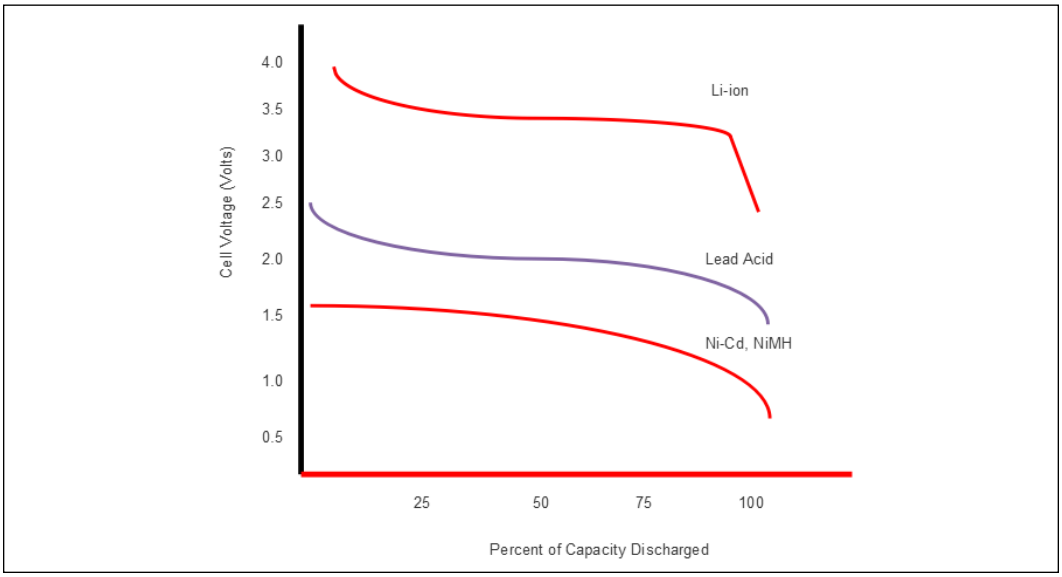


Figure 21: An example of relative discharge rates for various batteries. Li-ion provides a nearly constant voltage over its life but has a precipitous drop towards the end of its storage capacity.

The graph also illustrates that lead acid and Ni-Cd have less voltage potential and a curvilinear degradation in power that can be computed more reliably. The trailing slopes are also indicative of Peukert's capacity.

Temperature greatly affects battery life, and specifically the electro-active carriers in a cell. As temperature increases, the internal resistance of a battery decreases when being discharged. Even when batteries are stored, they can self-discharge, which impacts the total lifetime of a battery.

A Ragone plot is a useful way to show the relationship between energy storage systems when trading off energy capacity and power handling. It is a log-based scale where the energy density (Wh/kg) of a power source is plotted against the power density (W/kg). This relationship shows devices that tend to have a greater lifetime (batteries) versus devices that tend to store more energy (supercapacitors):

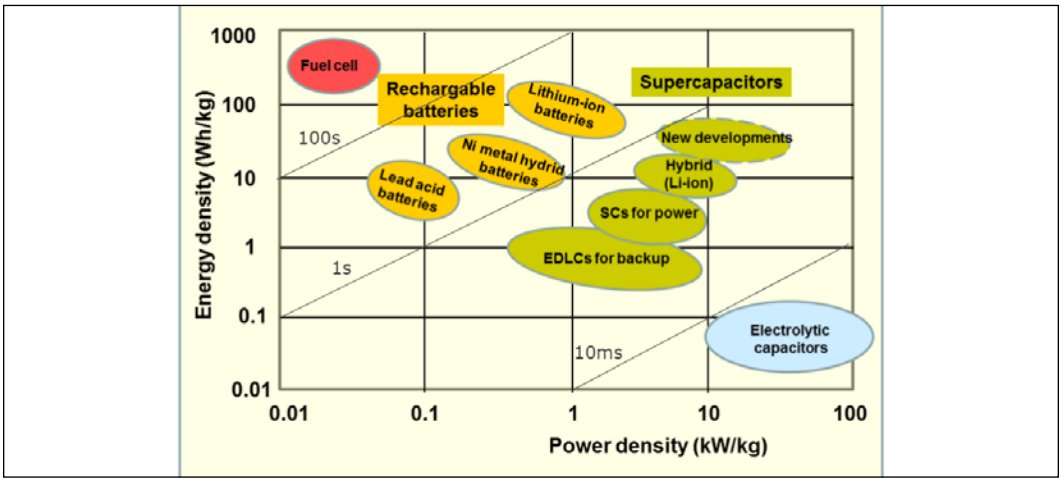


Figure 22: A Ragone plot depicting capacitor, supercapacitors, battery, and fuel cell differences in energy capacity versus lifetime supply of energy

Batteries like Li-ion have higher energy density and discharge rates than nickel-cadmium and nickel-hydride batteries. Capacitors produce very high power densities but have relatively weak energy density. Note the plot is log-based and also shows the discharge time for various storage systems.

Batteries

Typically, a **lithium-ion** (**Li-ion**) battery is the standard form of power in mobile devices due to its energy density. In such a battery, lithium ions physically move from the negative electrode to the positive. During recharge, the ions move back to the negative region. This is known as an *ionic movement*.

Batteries also develop memories with many charge-discharge cycles. This capacity loss is expressed as a measure of the initial capacity (for example, 30% loss after 1,000 cycles). This degradation is almost directly correlated to environmental temperature, and loss will increase in a high-temperature environment. Therefore, it is imperative that the architects manage thermals in a constrained environment if lithium-ion is to be used.

Another factor in battery life is self-discharge. When unwanted chemical reactions occur in a battery cell, energy will be lost. The rate of loss depends on chemistry and temperature. Typically, a Li-ion can last for 10 years (at ~2% loss per month), while an alkaline battery will only last 5 years (15% to 20% loss per month).

Supercapacitors

Supercapacitors (or supercaps) store energy at significantly higher volumes than typical capacitors. A typical capacitor will have an energy density of somewhere between 0.01 watt-hours/kg. A supercap has an energy density of 1 to 10 watt-hours/kg, thus placing them closer to the energy density of a battery that can be on the order of 200 watt-hours/kg. Like a capacitor, energy is stored electrostatically on a plate and doesn't involve the chemical transfer of energy like a battery. Usually, supercaps are made of fairly exotic materials like graphene, which can impact on overall cost. Supercaps have the advantage of charging to their full potential in seconds, whereas Li-ion batteries will charge within minutes to approximately 80%, but require a trickle current to safely get higher.

Additionally, supercaps can't be overcharged, whereas Li-ion can be overcharged and can result in serious safety concerns. Supercaps come in two forms:

- **Electric double-layer capacitors** (**EDLC**): Use an activated carbon electrode and store energy electrostatically
- **Pseudocapacitors**: Use a transition metal oxide and electrochemical charge transfer

Supercaps have an advantage over batteries in predicting the remaining time power will be available. The remaining energy can be predicted from the terminal voltage, which changes over time. Li-ion batteries have a flat energy profile from fully charged to discharged, thus rendering time estimation difficult. Since the voltage profile of a supercap changes over time, a DC-DC converter is needed to compensate for wide variations in voltage.

In general, the main issues with supercaps or capacitors are the leakage current and the cost. As can be seen in the table later in this section, supercaps have their place. One will often see them in a hybrid solution with regular batteries to provide instantaneous power (for example, electric vehicle acceleration), while the battery supply sustains running power.

Radioactive power sources

A radioactive source with a high energy density (10^5kJ/cm^3) can generate thermal energy due to the kinetic energy of emitted particles. Sources such as cesium-137 have a half-life of 30 years and a power capacity of 0.015 W/gm. This method can generate power in the watt-to-kW range but isn't practical in low-power sensor levels for IoT deployments. Space vehicles have used this technology for decades. Promising developments using MEMS piezoelectronics that capture electrons and force a micro-armature to move can create mechanical energy that may be harvested. A secondary effect of radioactive decay is the relatively weak power density profile. A radiation source with a long half-life will have weaker power density. Thus, they are suitable for bulk-charging supercaps to provide momentary energy when needed. The final issue with radioactive sources is the significant weight of lead shielding required. Cesium-137 requires 80 mm/W of shielding, which can add significant cost and weight to an IoT sensor.



Examples of radioactive power sources include space vehicles such as the Curiosity Mars Rover and New Horizon spaceship.

Energy storage summary and other forms of power

Choosing the correct power source is critical, as mentioned earlier. The following table provides a summary comparison of different components in a system to be considered when picking the correct power source:

Category	**Li-ion Battery**	**Supercap**
Energy density	200 Wh/kg	8-10 Wh/kg
Charge-discharge cycles	Capacity drops after 100 to 1,000 cycles	Nearly infinite
Charge-discharge time	1 to 10 hours	Milliseconds to seconds
Operational temperature	-20°C to +65°C	-40°C to +85°C
Operational voltage	1.2V to 4.2V	1V to 3V
Power delivery	Constant voltage over time	Linear or exponential decay
Charge rate	(Very slow) 40 C/x	(Very fast) 1,500 C/x
Operational life	0.5 to 5 years	5 to 20 years

Form factor	Very small	Large
Cost ($/kWh)	Low ($250 to $1,000)	High ($10,000)

Summary

This chapter summarized several different sensors and endpoints used in IoT deployments. The IoT isn't about simply connecting a device to the Internet. While that is a key component, the essence of IoT is connecting the analog world to the digital. Essentially, things and devices previously unconnected now have the opportunity to collect information and communicate it to other devices. This is powerful because data that had never been captured now has value. The ability to sense the environment will lead to more efficiencies, revenue streams, and value for customers. Sensing allows for smart cities, predictive maintenance, tracking assets, and analyzing the hidden meaning in massive aggregates of data. Powering such systems is also critical and must be understood by architects. A poorly designed system can lead to too short a battery life, which will end in substantial costs to remediate.

The next chapter will bridge the endpoints to the Internet through non-IP communication. We will explore various wireless personal area networks and their various traits and functions that are prevalent in the IoT space.